

17AUS4

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Problem

Find all pairs (a, b) of non-negative integers such that $2017^a = b^6 - 32b + 1$.

Solution. Observe that b is even. So $64 \mid 2017^a - 1 \implies 2 \mid 2017^{a-1} + 2017^{a-2} + \dots + 1 \implies 2 \mid a$.

For $b > 4$ we have $(b^3 - 1)^2 < b^6 - 32b + 1 < (b^3)^2$, so it is impossible for $b^6 - 32b + 1$ to be a perfect square.

For $b = 4$, $4^6 - 32 \times 4 + 1 = 129$, not a perfect power of 2017.

For $b = 2$, $2^6 - 32 \times 2 + 1 = 1 = 2017^0$.

For $b = 0$, $0^6 - 32 \times 0 + 1 = 1 = 2017^0$.

So the only pairs (a, b) that work are $\boxed{(0, 0), (0, 2)}$.